Start with the synchronous buck

Two controlled switches set the switching-node voltage. The output filter averages it.

Read from left to right: Vin supplies energy, the switches select a voltage at SW, and L + Co smooth it into Vo. The load is the device being powered.

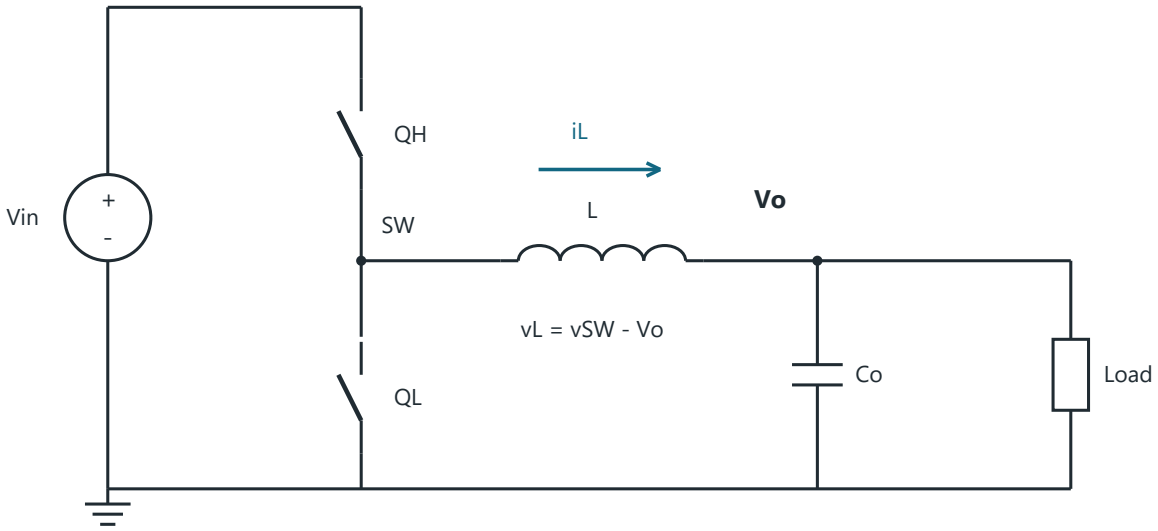


Figure 1. Ideal synchronous buck. QH = high-side switch; QL = low-side synchronous rectifier. All voltages reference the bottom rail.

State	Switching node	Inductor behavior
QH ON; QL OFF	vSW = Vin	vL = Vin - Vo; current rises
QH OFF; QL ON	vSW ≈ 0	vL = -Vo; current falls

$$D = t_{ON} / T_s \qquad f_s = 1 / T_s \qquad V_o = D\, V_{in}$$

Freewheeling means current circulates through QL, L and the output after QH turns off. L releases stored energy, so current falls gradually instead of stopping. Using QL usually loses less power than using a diode.

Dead time: briefly turn both switches off at a transition. Turning QH and QL on together shorts the input. The output capacitor supplies or absorbs the difference between inductor and load current.

A buck example, step by step

Use the ordinary circuit on the previous page before moving to the SSCB.

Imagine a 48 V source supplying a 12 V load. Turning QH on and off applies pulses at SW. We vary their duration; we do not ask QH to produce a continuous 12 V.

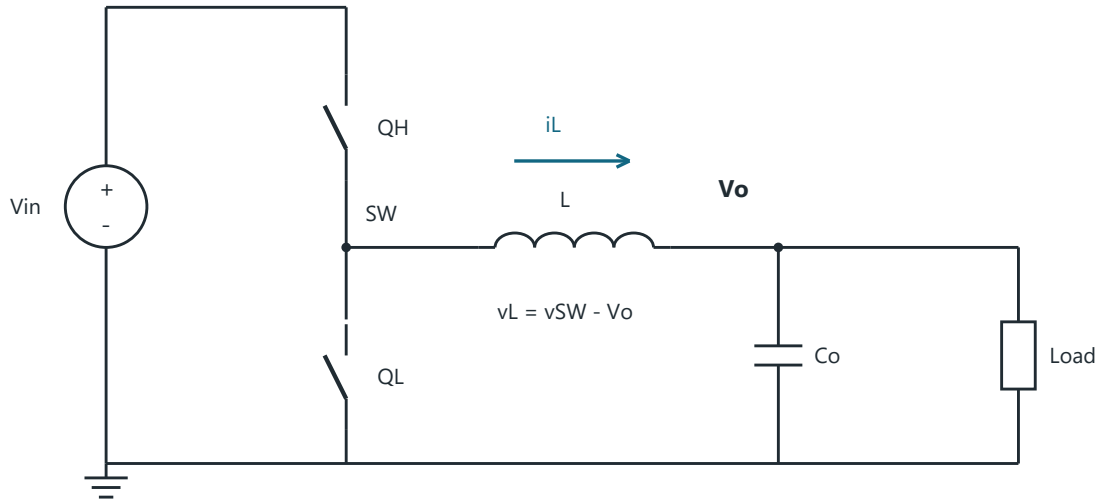


Figure 1a. Same circuit as Figure 1. Example values: Vin = 48 V, Vo = 12 V, fs = 200 kHz. These are teaching values, not the prototype inductor design.

D = 12 / 48 = 0.25 Ts = 1 / fs = 5 µs

QH conducts for 1.25 µs of each 5 µs cycle. Then QL carries current for about 3.75 µs, ignoring dead time. SW alternates between 48 V and 0 V; Vo remains near 12 V because L and Co smooth the pulses.

Why current rises, then falls

When QH is ON: vL = 48 - 12 = +36 V. When QL is ON: vL = 0 - 12 = -12 V. A larger positive voltage acts for less time; a smaller negative voltage acts for longer.

For L = 10 µH: rise = 4.5 A; fall = 4.5 A each cycle

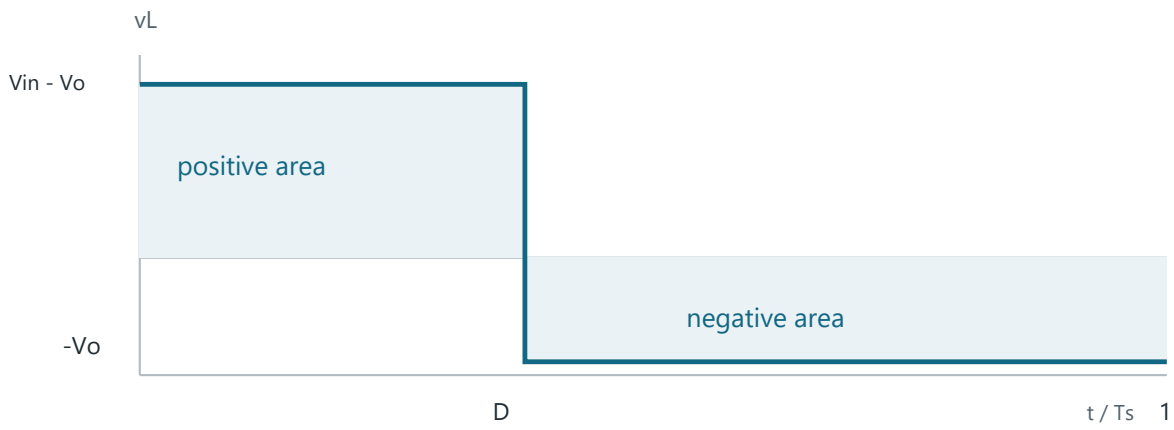
Calculation: rise = 36 × 1.25 / 10 = 4.5 A; fall magnitude = 12 × 3.75 / 10 = 4.5 A (µs and µH cancel). These are ripple changes, not the average load current. The next page explains why they must match.

Two balance laws

Steady state means each switching cycle returns stored energy and charge to the same values.

Inductor: zero net volt-seconds

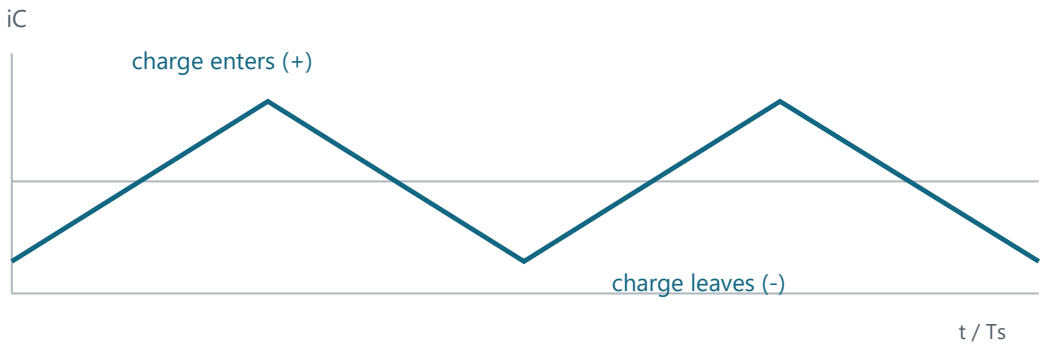
Reference: L in Figure 1; vL is positive from SW toward Vo.



$$D(Vin - Vo) + (1 - D)(-Vo) = 0 \quad \rightarrow \quad Vo = D\ Vin$$

An inductor changes current gradually: $\Delta iL = vL\ \Delta t / L$. Positive voltage makes current rise; negative voltage makes it fall. Equal positive and negative areas return current to its starting value each cycle.

Capacitor: zero net charge



$$\Delta Q = \int iC\ dt = 0 \text{ per cycle} \qquad \Delta VC = \Delta Q / C$$

Current entering a capacitor adds charge and raises voltage; current leaving removes charge and lowers it. Co carries iL - Iload. Zero net charge means its voltage repeats each cycle instead of drifting. $\int iC\ dt$ means the signed area under the current curve.

Interleaving and regulation

Parallel phases share load current. Shifting their ripple can reduce what reaches the output capacitor.

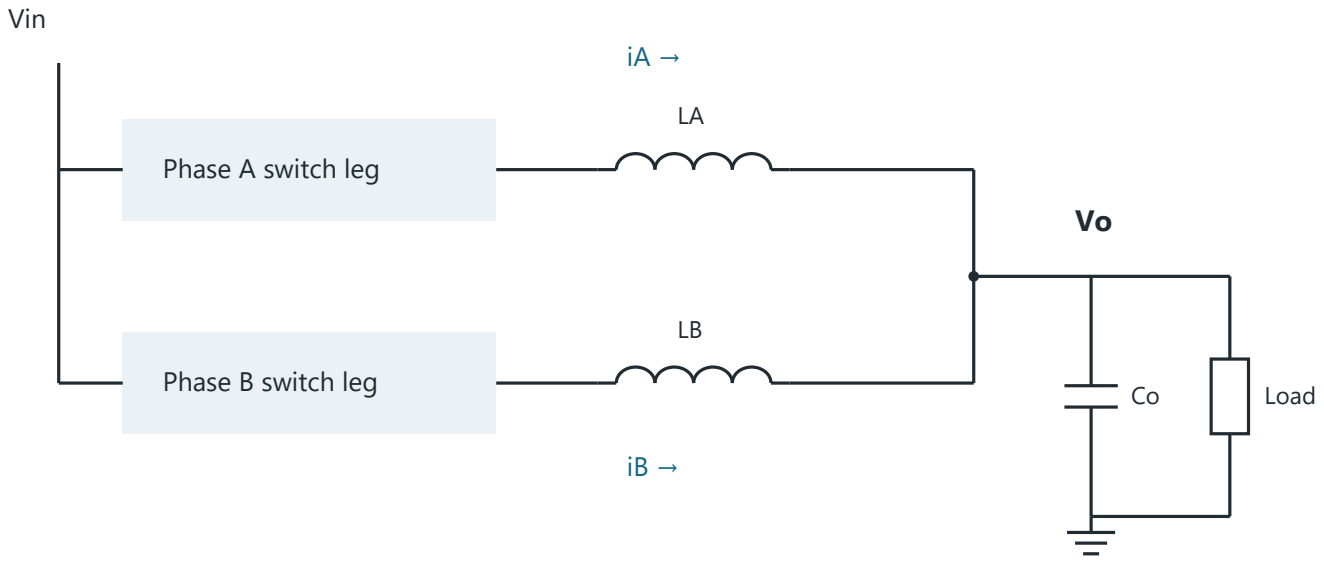


Figure 2. Ordinary two-phase buck, shown as switch-leg blocks. Both inductors feed the same output; iCo = iA + iB - Iload.

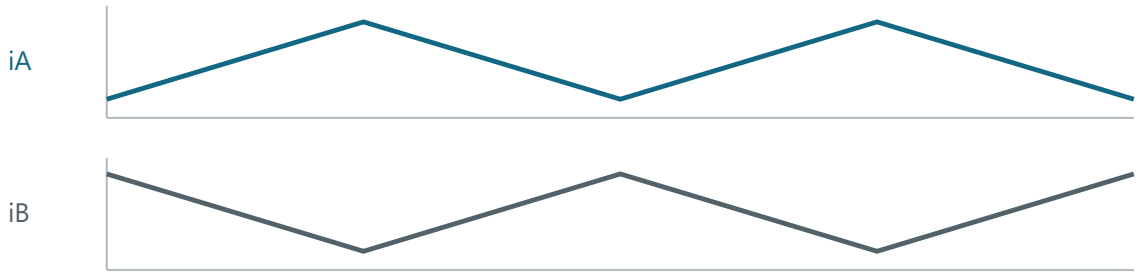
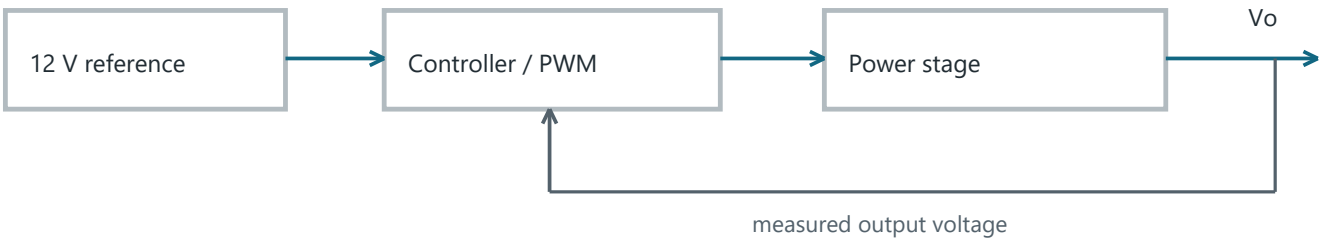


Illustration: matched phases, 180° shift, 50% inductor energizing duty.

Current sharing: ideally each phase carries Iload/2. Ripple cancellation depends on duty, matching and phase shift; it is not perfect at every operating point.



Regulation: feedback adjusts duty when input or load changes. Self-balancing of phase currents or flying-capacitor voltages does not replace this voltage-control loop.

What interleaving actually changes

Sharing the load and canceling ripple are related benefits, but they are different ideas.

First, share the average current

For an 80 A load, two balanced phases each carry 40 A on average. Neither phase is simply off for half the time: each inductor current can continue flowing through its synchronous rectifier.

Then, shift the ripple in time

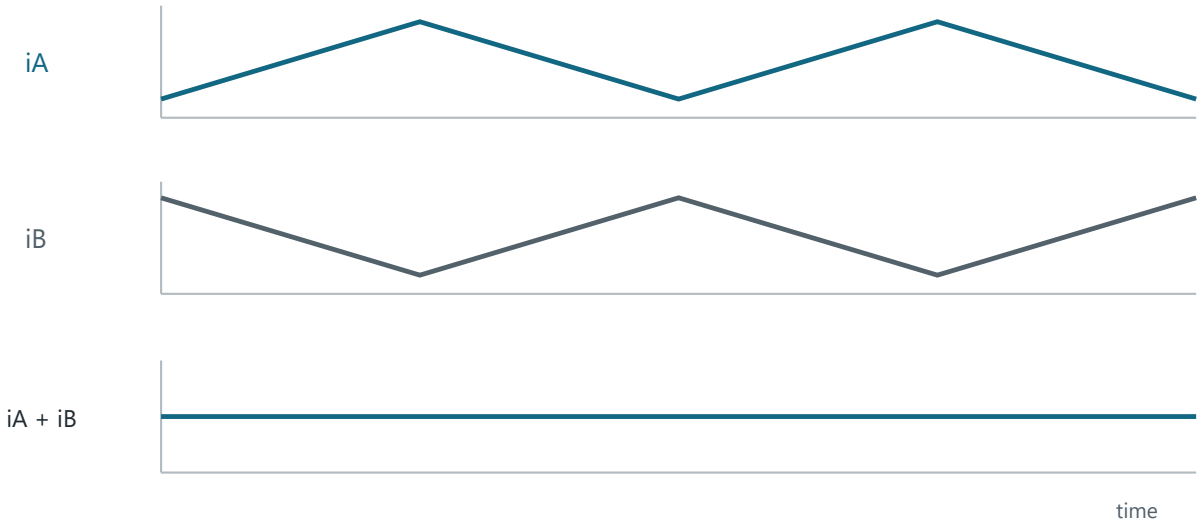


Figure 2a. Ideal matched phases at 50% energizing duty. As one ripple rises, the other falls. Their sum is constant in this special example. At other duties, residual ripple remains.

The output capacitor only handles iA + iB - Iload. Less ripple in the sum means less alternating capacitor current and usually less output-voltage ripple. Interleaving does not remove the need for Co.

Regulation answers a different question

A fixed 4:1 converter produces 12 V only at 48 V input: 40 V gives 10 V and 60 V gives 15 V. A regulated converter measures Vo and changes duty to keep it near 12 V. This is why a feedback controller is needed.

PWM means pulse-width modulation: changing switch ON time while repeating at the switching frequency. An error signal compares measured Vo with the desired voltage and guides the duty adjustment.

The SSCB circuit

The same buck principles apply, but C1 and C2 provide additional energy-transfer paths.

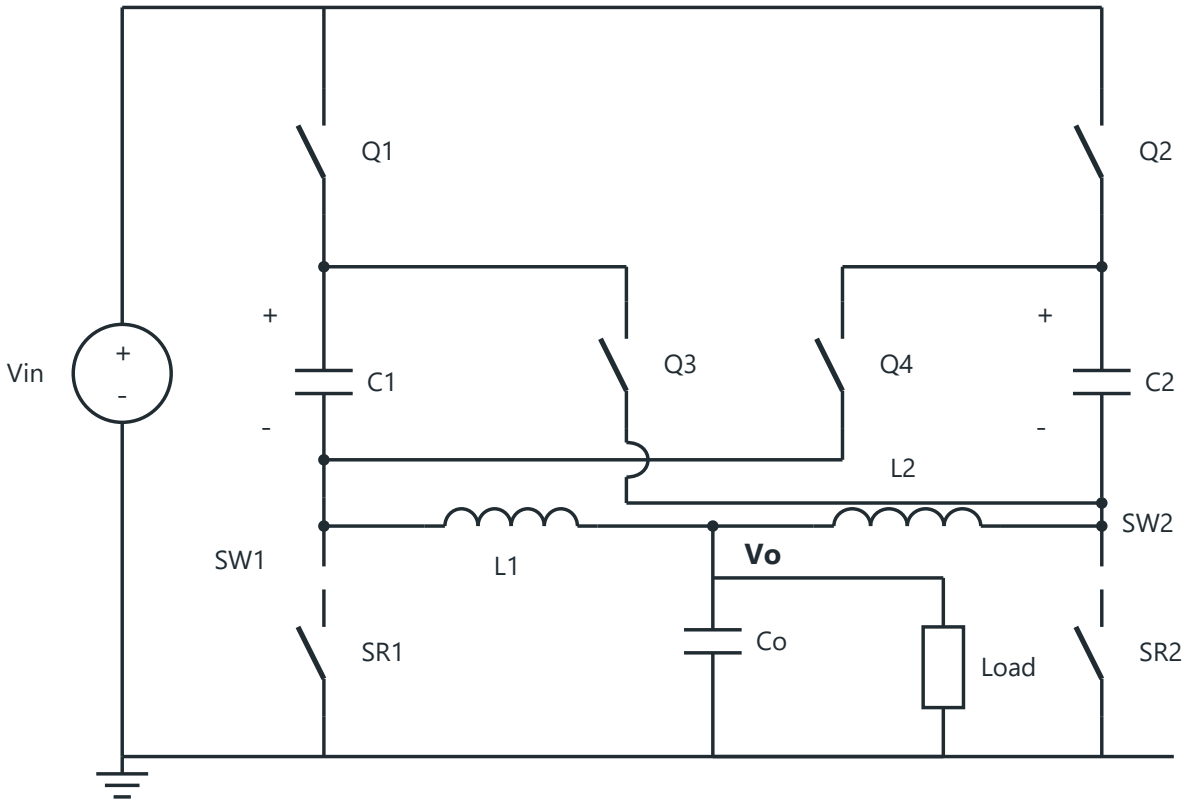


Figure 3. Reference circuit, redrawn from [1, Fig. 3]. Dots mean connected wires; the curved crossing is not connected. Capacitor voltage is top minus bottom. Both inductor currents flow toward Vo. Switch symbols show connectivity, not one operating state.

$$V_{C1} = V_{C2} \approx V_{in} / 2 \qquad \text{average } i_{L1} = \text{average } i_{L2} \approx I_{load} / 2$$

Component	Role / ideal blocking stress
Q1, Q2	Connect input to the flying-capacitor paths; about Vin/2 stress
Q3, Q4	Cross-connect each capacitor to the opposite phase; up to Vin stress
SR1, SR2	Freewheel the phase currents; about Vin/2 stress

Reading the equations: VC1 and VC2 are the voltages across C1/C2; IL1 and IL2 mean cycle-average currents. At 48 V input and 80 A load, the ideal balance values are 24 V per capacitor and 40 A per phase.

Follow one phase through two paths

A flying capacitor transfers charge between changing switch connections. It is not the output smoothing capacitor.

Path A: the input charges C1 and feeds L1

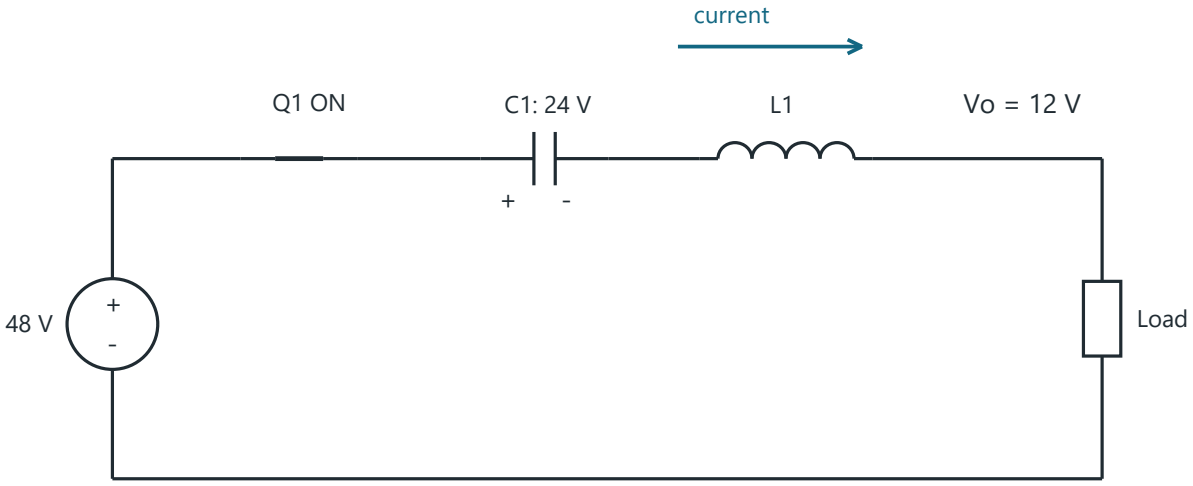


Figure 3a. Q1 path from Figure 3; Co and the other phase are omitted, and Vo ≈ 12 V is assumed. C1 drops 24 V, leaving SW1 near 24 V. Current enters the positive C1 terminal: C1 charges.

L1 does not see the whole 24 V: its other terminal is already at 12 V. Therefore vL1 = 24 - 12 = +12 V, and current rises. The capacitor voltage changes only slightly during one interval.

Path B: C2 takes over as the energy source

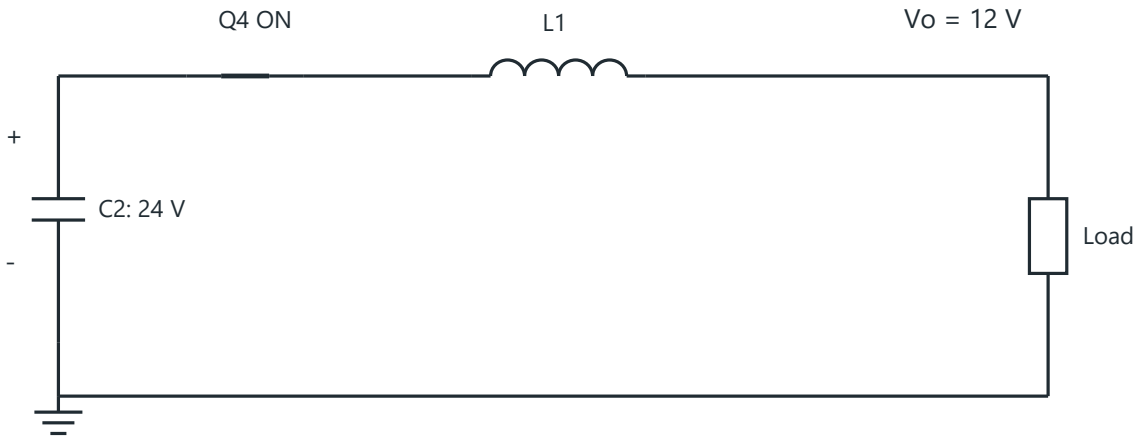


Figure 3b. C2 now supplies current through Q4 into L1. Its lower terminal returns through SR2 (shown as a wire). Current leaves the positive C2 terminal: C2 discharges. SW1 is still near 24 V.

The handover changes where energy comes from, while L1 keeps seeing +12 V. That is why two sequential switch intervals form one longer inductor-energizing interval. L2 uses the mirror-image paths.

A longer useful energizing interval

An inductor responds to its total applied volt-seconds, not to which switch supplied them.

Use d = ON time of each transfer switch / Ts throughout this report.

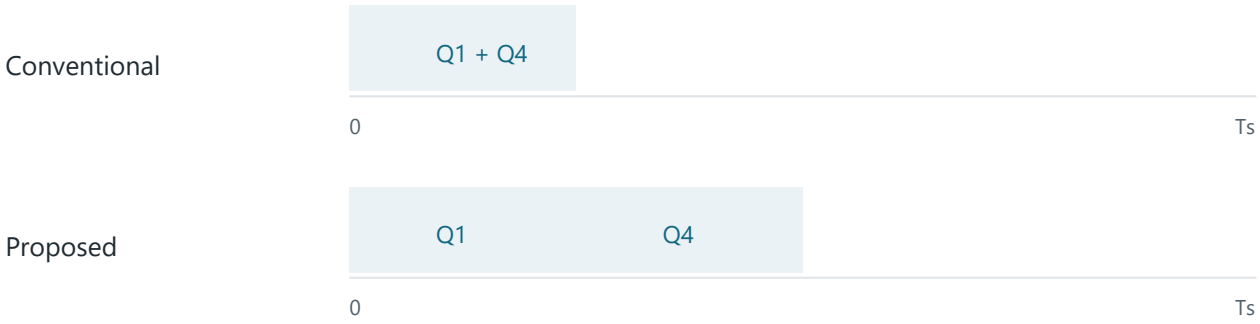


Figure 4. At the same d = 0.25, simultaneous paths give one 0.25Ts interval; sequential paths give two. SW1 stays near Vin/2 for 0.50Ts. This doubles its average voltage, not its pulse height.

	Conventional	Proposed
SW1 high fraction	d	2d
Conversion M = Vo/Vin	d/2; d < 1/2	d; d < 1/3
Maximum ideal M	1/4	1/3

$$2d(Vin/2 - Vo) + (1 - 2d)(-Vo) = 0 \quad \rightarrow \quad Vo = d\,Vin$$

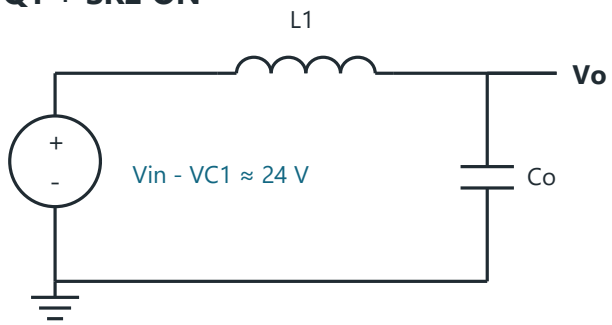
Input → 12 V	Conventional d	Proposed d
40 V	0.60: outside range	0.30
48 V	0.50: ideal boundary	0.25
60 V	0.40	0.20

Notation in the sources: [1] uses D = d. Slides 7-10 of [2] use D = 2d (total inductor energizing duty), so M = D/2 and D < 2/3. Slide 11 returns to the paper convention. The report keeps d consistent.

Four states at 48 V → 12 V

At d = 0.25, the ideal cycle divides into four equal intervals. Read against Figure 3.

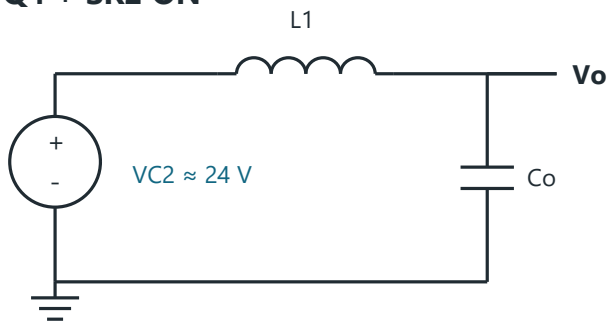
01 Q1 + SR2 ON



Vin sends current through Q1 and C1 into L1. C1 gains charge. SW1 is about 24 V, so L1 sees 24 - 12 = +12 V and its current rises.

L2 freewheels through SR2; its current falls.

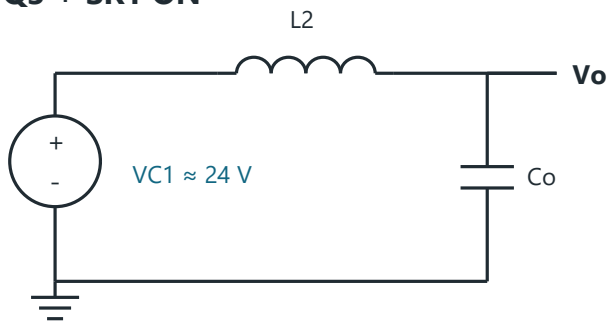
02 Q4 + SR2 ON



Q4 connects the charged C2 to L1. C2 now supplies energy and loses charge. SW1 stays near 24 V, so L1 current continues to rise.

L2 keeps freewheeling through SR2.

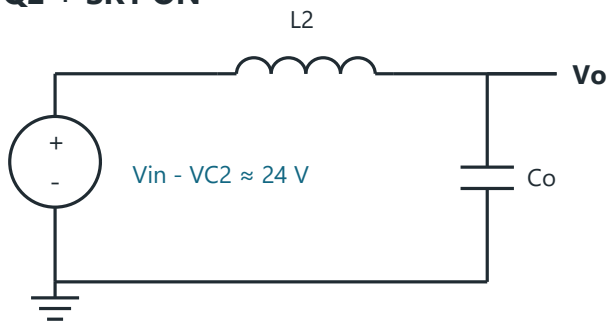
03 Q3 + SR1 ON



Q3 connects the charged C1 to L2. C1 supplies energy and loses charge. SW2 is near 24 V, so L2 current starts to rise.

L1 freewheels through SR1; its current falls.

04 Q2 + SR1 ON



Vin sends current through Q2 and C2 into L2. C2 regains charge. SW2 stays near 24 V, so L2 current continues to rise.

L1 keeps freewheeling through SR1.

Figure 5. Energized-phase equivalents; the source replaces its active switch/capacitor path. Other transfer switches are OFF. Energizing vL = +12 V; freewheeling vL = -12 V. Brief commutation is omitted. At 40 V, states overlap and these equal intervals no longer apply.

Why a tiny overlap is useful

Most switch transitions need dead time. This specific transfer between energy paths also needs a controlled overlap.

At the end of Path A, Q4 turns on just before Q1 turns off. For a few nanoseconds, Q1, Q4 and SR2 are ON together. Trace Figure 3: this briefly places C1 and C2 in series across Vin.

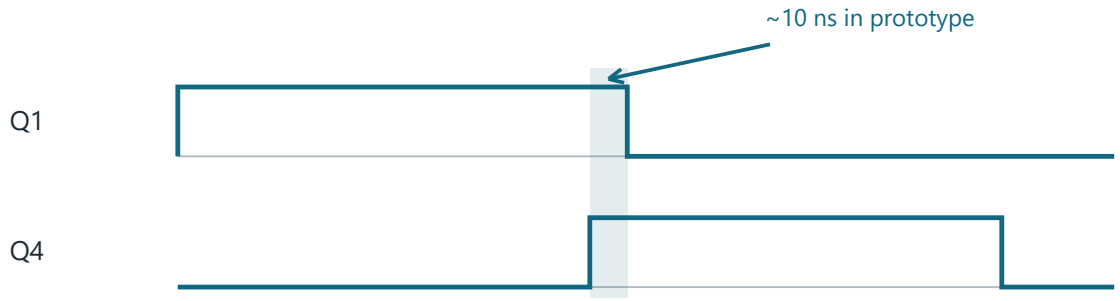


Figure 6. Q1/Q4 timing, not to scale. The Q3 → Q2 handover uses a corresponding overlap. The overlap is brief; it is not an extra long energy-delivery state.

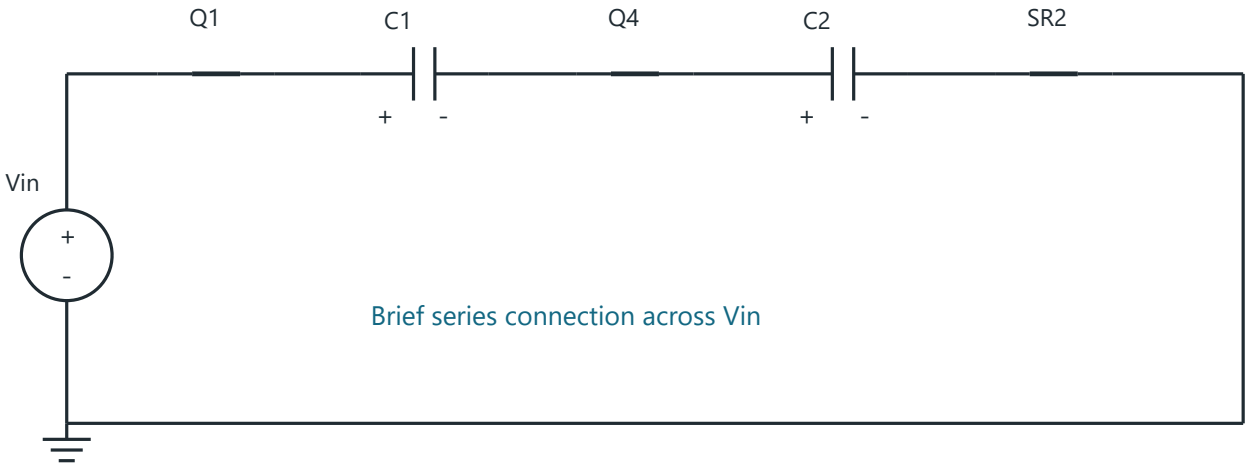


Figure 6a. Balancing-loop equivalent during overlap. Co and the inductor branches are omitted to make the series voltage relation clear. This is the same connectivity as Figure 3.

VC1 + VC2 ≈ Vin during the balancing connection

This connection constrains the sum of the two capacitor voltages. It does not, by itself, prove that each is Vin/2. The next page combines it with inductor volt-second balance to show equal sharing.

Do not copy the overlap to the QH/QL pair in an ordinary buck: that pair would short the input. Keep dead time at the required synchronous-rectifier transitions. Real overlap depends on drivers, device delays and parasitic impedance.

How the balance equations work

Each term is a voltage applied to an inductor, multiplied by the fraction of the cycle for which it is applied.

1. Add the two energizing contributions for L1

$$Vo = d(Vin - VC1) + dVC2$$

The first term is Path A through Q1; the second is Path B through Q4. During the rest of the cycle SW1 is near zero, so it contributes zero to its average. In steady state, average SW1 voltage equals Vo.

2. Write the mirror-image relation for L2

$$Vo = d(Vin - VC2) + dVC1$$

Both phases feed the same output, so their left sides are equal. Subtract the equations: $0 = 2d(VC2 - VC1)$. For nonzero duty, this requires $VC1 = VC2$. Combine with their voltage sum from the overlap:

$$VC1 = VC2 = Vin/2 \qquad \text{and therefore } Vo = d\,Vin$$

3. Balance charge to find the average phase currents

Over a cycle, each flying capacitor must regain the charge it gave away. With equal-duration intervals and small ripple, this gives $d\,IL1 = d\,IL2$. Thus the average phase currents are equal. Together they supply the load:

$$IL1 + IL2 = Iload \quad \rightarrow \quad IL1 = IL2 = Iload/2$$

At 80 A output, that is 40 A per phase. “Self-balancing” means these internal quantities settle around equal values in the intended steady-state operation. The output controller still sets d to maintain 12 V.

These relations assume continuous inductor current, small capacitor ripple, symmetric timing and idealized devices. They explain the operating point; they do not prove startup behavior or control-loop stability.

Hardware evidence and design lessons

The prototype demonstrates the modulation together with GaN switches, coupled magnetics and careful PCB layout.

Measured / reported [1]	Value
Input; regulated output	40-60 V; 12 V, up to 80 A (960 W)
Switching frequency	200 kHz for all six switches
48 V input efficiency	98.88% peak; 98.11% at full load
Footprint power density	787 W/in² (area-based, not volume-based)
Thermal test conditions	25 °C ambient; 400 LFM forced airflow

Three practical consequences

1. Lower voltage stress helps switching loss. For a simple constant-C estimate, $E = \frac{1}{2}CV^2$. Halving voltage quarters that energy term, not total converter loss. Use the device's actual output-capacitance energy data for design.

2. Shifted currents support coupled magnetics. Proper winding polarity gives negative coupling; phase fluxes partially cancel in the shared core. This can improve magnetic utilization, but does not provide electrical isolation.

3. Commutation layout matters. Parasitic overshoot follows $\Delta V = L_{loop}\, di/dt$. Keep the transfer-switch loops compact. The paper reports PCB loop contributions below 1 nH; device and interconnect parasitics still matter.

At 960 W: $P_{loss} = 960(1/0.9811 - 1) \approx 18.5$ W

At 40 V input, the reported hottest surface reaches about 88.3 °C under the stated cooling. This is a surface measurement, not semiconductor junction temperature.

Sources. [1] X. Xu and Q. Li, "Symmetric Series-Capacitor Buck in 48V-to-12V Regulated Conversion for High-Performance Server Boards," IEEE ECCE, 2024. Supplied paper: PDF pp. 3-8 (circuit, modulation, balance, layout); pp. 11-12 (results).
[2] X. Xu and Q. Li, presentation of the same title, Virginia Tech CPES, 25 Nov 2024. Supplied slides 4-11 (operation); 12-16 (hardware and conclusions). Circuit diagrams are redrawn; timing sketches are idealized.